(a) Production-stack validation

Pair products	**504 / 504**
Common-referenced	**504 / 504**
Ancillary missing	**0**
Ancillary problems	**0**
Baseline rank	**181 / 181**
Full GEOC readiness	**true**

Numerical diagnostics

East/north/up median norm
1.000000

Maximum norm error
1.19×10^{-7}

Baseline residual RMS
0.0155 m

(b) Portability tests

Windows production stack
Python 3.11.15 | stdlib fallback

Linux synthetic
Python 3.13.5 | matplotlib

Linux synthetic
Python 3.13.5 | stdlib

Linux synthetic
Python 3.13.5 | auto; Matplotlib healthy

Linux synthetic
Python 3.13.5 | auto; Matplotlib disabled